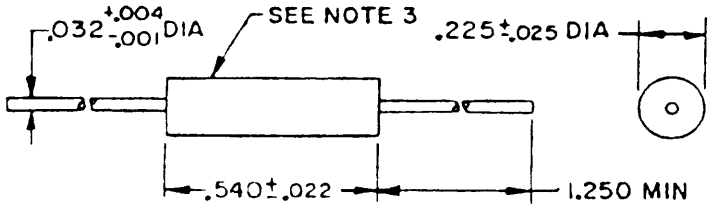
MILITARY SPECIFICATION SHEET

CAPACITORS, FIXED, CERAMIC DIELECTRIC (GENERAL PURPOSE), STYLE CK21

This specification is approved for use by all Departments and Agencies of the Department of Defense.

**NOT FOR NEW DESIGN
FOR REPLACEMENT PURPOSES ONLY**

The requirement for acquiring the capacitors described herein shall consist of this specification and the latest issue of MIL-C-11015.



Inches	mm
.001	0.03
.004	0.10
.022	0.56
.025	0.64
.032	0.81
.225	5.72
.540	13.72
1.250	31.75

NOTES:
1. Dimension are in inches.
2. Metric equivalents are given for general information only.
3. The beginning of the type designation shall be adjacent to the end having the inner-electrode terminal.

REQUIREMENTS:

Design and construction:
Dimensions and configuration: See above figure.
Case type: Tubular with axial leads.
Capacitance value: See table I.
Rated voltage: 500 V dc.
Rated temperature and voltage temperature limits: AX.
Capacitance tolerance: ±10 percent (K) or ± 20 percent (M) as shown in table I.
Rated temperature: -55°C to +85°C.

Dielectric withstanding voltage:
Dielectric: Method 301 of MIL-STD-202.
Test voltage: 1,250 V dc.
Body insulation: Shall be able to withstand 1,300 volts for 5 ±1 seconds between leads and three turns of AWG size 20 copper wire wrapped, closely-spaced, around the capacitor body at the approximate center.

Barometric pressure (reduced): Method 105 of MIL-STD-202. 0.82 inch of mercury (80,000 feet).
Test potential: 750 V dc.

Insulation resistance (IR): Method 302 of MIL-STD-202, condition B. 20,000 megohms, minimum.

Capacitance: Method 305 of MIL-STD-202.
Test frequency:
100 pF or less: 1 MHz ±1 kHz.
Over 100 pF: 1 kHz ±100 Hz.
Test voltage: 1.0 ±0.2 volt rms.

Ⓓ denotes changes

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TABLE I. Style CK21 characteristics.

Type designation 1/	Capacitance	Capacitance tolerance
	pF	
CK21AX120K	12	K
CK21AX150K	15	K
CK21AX180K	18	K
CK21AX220K	22	K
CK21AX270K	27	K
CK21AX330K	33	K
CK21AX390K	39	K
CK21AX470K	47	K
CK21AX560K	56	K
CK21AX680-	68.	K, M
CK21AX820K	82	K
CK21AX101-	100	K, M
CK21AX121K	120	K
CK21AX151-	150	K, M
CK21AX181K	180	K
CK21AX221-	220	K, M
CK21AX271K	270	K
CK21AX331-	330	K, M
CK21AX471-	470	K, M
CK21AX681-	680	K, M

1/ Where applicable, the complete type designation will include an additional symbol to indicate capacitance tolerance.

Dissipation factor (DF): 1.5 percent maximum.

(D) Shock, specified pulse: Not applicable.

Vibration, high frequency: Method 204 of MIL-STD-202, condition B (15 G's).

(D) Thermal shock and immersion:
(D) Thermal shock: Method 107 of MIL-STD-202, condition A.
Immersion: Method 104 of MIL-STD-202, condition B.
IR: 3,000 megohms, minimum.

Salt spray (corrosion): Not applicable.

Terminal strength: In accordance with MIL-C-11015, five pounds applied.

Moisture resistance: Method 106 of MIL-STD-202.
IR: 3,000 megohms, minimum.

Solderability: Method 208 of MIL-STD-202. Two terminals.

Resistance to soldering heat: Not applicable.

Voltage-temperature limits: In accordance with MIL-C-11015.

Life (at elevated ambient temperature): Method 108 of MIL-STD-202. 1,000 hours (for qualification); 250 hours (performance check); 750 hours (continuation test).
Test potential: 1,000 V dc.
IR: 500 megohms, minimum (at 85°C); 10,000 megohms, minimum (at 25°C).
DF: 3 percent, maximum (at 25°C).

Marking: In accordance with MIL-C-11015.

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Custodians:
Army - ER
Navy - EC
Air Force - 85

Review activities:
Army - EA, MI
Navy - AS, OS, SH
Air Force - 11, 17, 99
DLA - ES

User activities:
Navy - MC
Air Force - 19

Preparing activity:
Navy - EC

Agent:
DLA - ES

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